Explainable AI Diabetes Risk Assessment

Academic Research Prototype | Assessment Date: September 07, 2026 - 22:58:27

1. Patient Information

Patient Name: Sanjith Kumar	**Biological Sex:** Male
Age: 33 years	**Assessment Date:** September 07, 2026 - 22:58:27

2. Clinical Measurements

Pregnancies: 0	**Glucose:** 145 mg/dL
Blood Pressure: 74 mmHg	**Skin Thickness:** 22 mm
Insulin: 110 muU/mL	**BMI:** 31.2 kg/m²
Diabetes Pedigree Function: 0.52	*(Patient Age recorded in Patient Information above)*

3. Risk Assessment

Risk Classification: **ELEVATED / HIGHER RISK (Class 1)**
Estimated Probability: 57.08% | **Decision Threshold:** 50.0%
Evaluated Architecture: XGBoost (Calibrated) (Holdout ROC-AUC: 0.8198)

4. Top Influential Factors

Local SHAP (SHapley Additive exPlanations) features ranked by contribution to this prediction:

Rank	Clinical Parameter	Patient Value	SHAP Attribution	Directional Impact on Model
1	**Glucose**	145.0	+0.5185	Pushes model prediction toward diabetes-positive class
2	**BMI**	31.2	+0.2842	Pushes model prediction toward diabetes-positive class
3	**DiabetesPedigreeFunction**	0.52	+0.2750	Pushes model prediction toward diabetes-positive class
4	**Age**	33.0	+0.1611	Pushes model prediction toward diabetes-positive class
5	**Pregnancies**	0.0	-0.1332	Pushes model prediction toward diabetes-negative class

Interpretability Note: *SHAP values describe how individual features influenced this model's prediction for this patient. They do not establish biological causation or clinical importance.*

5. Personalized Model-Based Summary

Based on this assessment, the model's strongest upward contributors were Glucose and BMI. These factors had positive local SHAP contributions and therefore pushed the model's prediction toward the diabetes-positive class. The following recommendations are educational guidance related to these model contributors.

Highest-Priority Areas for this Model Prediction:

1. Blood Glucose (Glucose) — Patient Value: 145.0 | Positive SHAP Contribution: +0.5185 [*Modifiable factor*]
Why it matters to the model: Glucose is one of the factors contributing most strongly to the model's estimated risk for this assessment (positive SHAP contribution of +0.5185).

2. Body Mass Index (BMI) (BMI) — Patient Value: 31.2 | Positive SHAP Contribution: +0.2842 [*Modifiable factor*]
Why it matters to the model: BMI contributed toward a higher model-estimated risk in this assessment (positive SHAP contribution of +0.2842).

3. Diabetes Pedigree Function (DiabetesPedigreeFunction) — Patient Value: 0.52 | Positive SHAP Contribution: +0.2750 [*Non-modifiable factor*]
Why it matters to the model: DiabetesPedigreeFunction contributed toward the model's estimated risk. This feature represents family-history-related genetic information and is not directly modifiable (SHAP +0.2750).

4. Age (Age) — Patient Value: 33.0 | Positive SHAP Contribution: +0.1611 [*Non-modifiable factor*]
Why it matters to the model: Age is a non-modifiable factor that contributed toward the model's estimated risk (positive SHAP contribution of +0.1611).

Factors That Lowered Model-Estimated Risk:

- Pregnancy History lowered the model's predicted risk for this particular assessment (SHAP -0.1332).
- Serum Insulin lowered the model's predicted risk for this particular assessment (SHAP -0.1297).
- Triceps Skinfold Thickness lowered the model's predicted risk for this particular assessment (SHAP -0.0383).

6. Personalized Educational Guidance & Wellness Recommendations

Educational Guidance for Blood Glucose:

- Discuss appropriate blood-glucose testing and routine monitoring with a qualified healthcare professional.
- If not recently evaluated, discuss fasting plasma glucose and/or glycated hemoglobin (HbA1c) screening with a physician.
- Encourage balanced meals emphasizing vegetables, high-fiber foods, whole grains, and appropriate lean protein sources.
- Limit excessive intake of refined carbohydrates, confectioneries, and sugar-sweetened beverages.
- Engage in regular physical activity appropriate for your baseline fitness and health status.
- Clinical Notice: Do not independently initiate or alter any medication, insulin therapy, or treatment regimen without physician oversight.

Educational Guidance for Body Mass Index (BMI):

- Encourage sustainable, nutritionally balanced eating patterns rather than restrictive or crash diets.
- Maintain regular physical activity appropriate to personal fitness level, joint health, and medical history.
- If weight management is clinically appropriate, discuss realistic and individualized goals with a healthcare professional or registered dietitian.
- Clinical Notice: The system does not promise that altering body weight will reduce the computational prediction by a specific amount.

Educational Guidance for Diabetes Pedigree Function:

- Maintain active awareness of family medical history regarding diabetes and cardiovascular conditions.
- Share relevant family metabolic history during clinical discussions with your primary healthcare provider.
- Proactively maintain healthy lifestyle habits (nutrition and physical exercise) to optimize metabolic resilience.

Educational Guidance for Age:

- Maintain regular age-appropriate health screenings and cardiovascular wellness evaluations.
- Engage in consistent, low-impact physical activity suited for your age bracket and mobility.
- Maintain a balanced, nutrient-dense diet to support healthy metabolic function.
- Discuss regular clinical diabetes screening with a healthcare professional during routine check-ups.
- Clinical Notice: Age cannot be altered, and this guidance focuses purely on healthy aging and age-appropriate preventive vigilance.

7. General Preventive Wellness Guidance

- Given the elevated computational risk score, discuss confirmatory diagnostic laboratory testing (such as fasting glucose and HbA1c) with a licensed physician.
- Adopt balanced dietary patterns emphasizing complex carbohydrates, vegetables, and whole foods.
- Ensure adequate dietary fiber intake to support healthy metabolic and digestive functioning.
- Prioritize lean protein sources and healthy unsaturated fats across daily meals.
- Limit excessive consumption of refined sugars, processed carbohydrates, and sugar-sweetened beverages.
- Maintain regular physical activity (targeting at least 150 minutes of moderate aerobic activity weekly as tolerated).
- Engage in routine preventive cardiovascular and metabolic health screenings with a healthcare provider.
- Consult qualified healthcare professionals before undertaking major dietary, physical, or lifestyle changes.

8. AI Natural Language Synthesis

Computational prediction synthesized from model feature weights.

9. Important Limitations & Disclaimer

Clinical Limitations: This assessment is produced by an academic machine learning research prototype trained on the historical PIMA Indians Diabetes Dataset. It is not an FDA-cleared or CE-marked diagnostic medical device. Prediction scores describe statistical pattern associations and should never be interpreted as clinical certainty. Patients must not initiate, alter, or discontinue any medications or treatments without direct consultation with a qualified medical professional.

MANDATORY MEDICAL NOTICE: This report is generated by an AI-based research prototype for educational purposes. It is not a medical diagnosis and should not replace consultation with a qualified healthcare professional.